

Eshaan Marocha

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EDUCATION

University of Toronto

Sep. 2024 – Apr. 2029 (Expected)

Bachelor of Applied Science – Electrical Engineering + PEY Co-op

Toronto, ON

- GPA: 3.93/4.0 (90%+ average)
- Intended Minor: Artificial Intelligence
- Relevant Courses: Programming Fundamentals (C/C++), Digital Systems (Quartus Prime, ModelSim, Verilog), Introductory Electronics (LTspice), Hardware Design and Communications (Altium Designer)

SKILLS

Programming Languages: Python, C, C++, Verilog, SystemVerilog

Hardware & Embedded: RTL Design, FPGA Development, ASIC Design, Design Verification, Embedded Systems

AI & Software: Computer Vision, OpenCV, Google Gemini API, Firebase/Firestore, Node.js

Libraries: Pandas, NumPy, Matplotlib

Design & Simulation Tools: LTspice, Vivado, Quartus Prime, ModelSim, Altium Designer, KiCad, Fusion 360

WORK EXPERIENCE

Intelligent Sensory Microsystems Laboratory (University of Toronto)

May 2026 – Present

FPGA Design Intern

Toronto, ON

- Developed FPGA logic for a custom computational-imaging camera system, enabling precise control over when different regions of the camera sensor collect light.
- Designed and synthesized SystemVerilog FPGA logic in Vivado to synchronize laser projector scans with camera exposure timing at sub-microsecond speeds, allowing the system to separate light based on its path.
- Built a Python/OpenCV-to-FPGA communication path with DDR3 memory control, allowing the camera to use captured images to decide which sensor regions should collect light next during runtime.

Excel Drafting Inc.

May 2025 – Aug. 2025

Engineering Intern

Acton, ON

- Created company-wide macros for automatic project property updates, reducing manual editing time by 93%.
- Automated bill of material formatting using a Python script, eliminating formatting errors.

PROJECTS

Muscle Controlled Robotic Arm

Mar. 2026 – Apr. 2026

- Built a 5-degree-of-freedom robotic arm that translates human muscle flexion into real-time motion, using an electromyography sensor to detect electrical activity from muscle activation.
- Designed and 3D-modelled the full robotic arm from scratch in Fusion 360, creating custom mechanical structures for joint rotation and gripper compression.
- Fabricated and assembled the full mechanical system by 3D printing all arm components, integrating servo motors, and wiring power and signal lines.
- Developed embedded C software on a DE1-SoC running a Nios V processor to detect muscle flexes, switch between control modes, and generate pulse-width modulation signals to drive five servos in real time.
- Debugged hardware integration issues including an unusable analog-to-digital converter port, wire-management failures, and a burnt out gripper servo, enabling a working final demo within a 3-week build timeline.

Flexible HF Radio Transceiver

Jan. 2026 – Apr. 2026

- Designed a power amplifier and filter for a modular high frequency radio transceiver, translating an interface/specification document into schematics, a PCB, and a test plan.
- Used circuit simulation (LTSpice) to iterate on the schematic design and validate functionality.
- Translated the schematic into an Altium PCB layout to be fabricated and assembled.
- Tested the assembled PCB using Python test automation tools to validate the PCB against requirements.

Ray Tracing ASIC

Feb. 2026

- Designed a hardware software pipeline that turns 3D models into rendered images using ray tracing, where each pixel's color is computed by finding the first surface its corresponding ray hits in the scene.
- Preprocessed 3D models into a compact 3D scene representation (solid vs empty) and streamed this data into an Application-Specific Integrated Circuit (ASIC) through a dedicated scene-loading interface.
- Designed the ASIC to accept rays and repeatedly step through a scene to find the first hit location using pipelined compute stages resulting in an estimated throughput of 1.8 million rays per second on a $32 \times 32 \times 32$ scene.
- Integrated a software rendering application and GUI that sends one ray per pixel to the chip, then converts returned hits into final pixel colors using lighting calculations, shadows, and multiple light sources for realistic images.
- Made power, performance, and area (PPA) aware design decisions by using compact encodings, fixed-point arithmetic rather than floating point, and bounding worst-case work with a maximum step limit.
- Validated manufacturability by synthesizing the design in TinyTapeout producing a fabrication-ready netlist.
- Verified correctness at both the module and full-chip level using reference models that compute expected results, combining directed edge-case tests with large randomized regressions (10,000+ inputs) and repeatable seeds.

Cupid Glasses (Make U of T 2026 Hackathon)

Feb. 2026

- Developed wearable smart glasses that use a camera to identify who you are speaking with and display real-time, personalized icebreakers on an LCD, winning 1st Place and Best Use of Google Gemini API at MakeUofT 2026.
- Built a Flutter mobile app for creating and managing profiles of people you speak to, storing profile data (name, photo, interests) in Firebase so the backend can retrieve context instantly.
- Streamed live video from an ESP32-CAM attached to the glasses to a Python backend over a local WiFi network to enable server-side data processing.
- Processed incoming video frames with OpenCV and used the DeepFace recognition model to match the person in the live feed to the saved profile photo from the mobile app.
- Ran emotion inference on the video stream using Presage, to identify the emotion of the person you are speaking to
- Queried Firebase to pull the matched person's interests, then sent interests + live emotion context to the Google Gemini API to generate tailored icebreakers to be displayed on the LCD.

FPGA Drum Pad

Nov. 2025

- Built an electronic drum controller by programming a reconfigurable chip (FPGA), allowing drum sounds to be played when buttons are pressed.
- Programmed a sound engine on the FPGA by writing hardware logic in Verilog to read drum samples from on-chip memory and mix multiple sounds at once, enabling smooth, layered drum hits.
- Implemented a physical breadboard-based control interface by wiring 12 drum pads, a volume knob, and action buttons (play, record, etc.) to the FPGA via GPIO, resulting in an intuitive hardware front panel for the user.
- Designed and 3D-modelled a custom enclosure and button caps using Fusion 360 and 3D-printed them, creating an ergonomic playing surface.
- Verified and refined the design by simulating modules in ModelSim, synthesizing in Quartus, and iteratively testing on hardware, ensuring the final system behaved as intended.

Stock Trading Algorithm

May 2025 – Jun. 2025

- Developed a Python (+ NumPy and Pandas) based automated stock-trading strategy that scans all S&P 500 stocks and generates buy/sell signals.
- Evaluated the strategy by building a backtesting engine and testing it on real S&P 500 price data from Jun. 2024 to Jun. 2025, achieving a 24.8% compound annual growth rate (CAGR) over that period.
- Built a Python metrics layer to compute and report portfolio KPIs (e.g., drawdown, win rate, profit factor, CAGR, Sharpe ratio), enabling fast iteration and validation across backtests.
- Implemented visual performance analytics by plotting equity curves and benchmark comparisons to the S&P 500 with Matplotlib, making it easy to evaluate the strategy over time.

AWARDS

NSERC Undergraduate Student Research Award | *University of Toronto*

Mar. 2026

1st Place in Valentines Theme Make U of T 2026 | *IEEE U of T*

Feb. 2026

Best use of Gemini API Make U of T 2026 | *IEEE U of T*

Feb. 2026

3rd Place Overall Hack Without Borders 2025 | *Engineers Without Borders*

Mar. 2025

Walter Scott Guest Memorial Admission Scholarship | *University of Toronto*

Sep. 2024

Edward S. Rogers Sr. Admission Scholarship | *University of Toronto*

Sep. 2024